Crop Recommendation System Using Random Forest and Naive Bayes

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Crop Recommendation System Using Random Forest and Naive Bayes

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1. Abstract

Agriculture plays an important role in ensuring food security and supporting economic development. Selecting a suitable crop for cultivation is a challenging task because different crops require different soil and climatic conditions. Traditional crop selection methods mainly depend on farmers' experience and knowledge, which may not always provide accurate recommendations when environmental conditions change. This project proposes a Crop Recommendation System using Python and Machine Learning to recommend a suitable crop based on important agricultural parameters such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. The system uses a crop recommendation dataset and performs data preprocessing to prepare the data for training and testing. Two machine learning algorithms, Random Forest and Naive Bayes, are used to predict the most suitable crop for the given soil and environmental conditions. Random Forest combines multiple decision trees to provide reliable classification, while Naive Bayes uses probability-based classification to make predictions efficiently. The performance of both models is evaluated using accuracy, precision, recall, F1-score, and confusion matrix to compare their prediction performance.The main aim of this project is to provide a simple and useful system that can help farmers make better crop selection decisions based on soil and climatic conditions. The system can reduce guesswork, support efficient use of agricultural resources, improve crop selection, and contribute to more sustainable farming practices.

2. INTRODUCTION:

Agriculture is one of the most important sectors contributing to the economy and food production across the world. With the increasing demand for food due to rapid population growth, farmers are under constant pressure to maximize crop productivity while maintaining soil fertility and efficient resource utilization. Selecting the appropriate crop based on soil characteristics and climatic conditions is a critical factor that directly influences agricultural output. However, unpredictable weather patterns, changing environmental conditions, and limited knowledge about soil nutrients often make crop selection a difficult task. These challenges have motivated researchers to develop intelligent decision-support systems using machine learning techniques to assist farmers in selecting suitable crops. Machine learning has become an effective solution in precision agriculture because of its ability to analyze large volumes of agricultural data and identify complex relationships among various environmental factors. This project employs two widely used machine learning algorithms: Random Forest and Naive Bayes. Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions to improve classification accuracy while reducing overfitting. Naive Bayes is a probabilistic classifier based on Bayes' theorem that assumes independence among features, making it simple, fast, and effective for classification problems. The proposed Crop Recommendation System analyzes important agricultural parameters such as nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall to predict the most suitable crop for cultivation. By comparing the performance of Random Forest and Naive Bayes, the system identifies the algorithm that provides higher prediction accuracy and better recommendation capability, thereby supporting farmers in making data-driven agricultural decisions and improving overall crop productivity.

3. SOFTWARE REQUIREMENTS:

The proposed Crop Recommendation System is developed using Python and various machine learning and data analysis libraries in an integrated development environment. Data preprocessing, visualization, model training, and evaluation are performed using libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn. The Random Forest and Naive Bayes algorithms are implemented using the Scikit-learn library. The project is developed and tested using Jupyter Notebook or Google Colab, which provide an interactive environment for coding, visualization, and experimentation. Since both algorithms are computationally efficient and do not require extensive hardware resources, the system can be executed on a standard personal computer without requiring a dedicated GPU. The software

Category	Requirement
Operating System	Windows 10/11 or Ubuntu 20.04 and above
Programming Language	Python 3.10 or above
Machine Learning Framework	Scikit-learn
Supporting Libraries	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
Development Environment	Jupyter Notebook / VS Code / Google Colab
Processor	Intel Core i5 / i7 or equivalent
RAM	8 GB minimum, 16 GB recommended
GPU	Not required; CPU is sufficient.
Storage	20 GB free disk space

and hardware requirements for the proposed system are summarized below.

Signature of the Guide

Course Instructor